

# Hospital Readmission Decision Support

*Prediction to Prevention*

Presented by Chloe Prowse



# Business Problem

## **The Problem:**

Hospital readmissions can lead to higher costs, increased resource use, and additional burden on patients and care teams.

## **The Goal:**

Identify patients at higher risk of 30-day readmission so hospitals can provide more targeted support after discharge.

## **Primary Stakeholders:**

Hospital administrators and care coordination/discharge planning teams.

## **Key Question:**

Which patients are most likely to be readmitted, and how can hospitals act on that risk?



# FACE Framework

**Frame:**

Identify high-risk patients with diabetes for targeted discharge interventions to reduce 30-day readmissions, improve patient care, lower healthcare costs, and better allocate hospital resources.

**Analyze:**

Analyze the UCI Diabetes dataset, develop and evaluate predictive models, build a RAG system using an LLM with CMS and AHRQ guidance, and conduct a cost/value analysis.

**Communicate:**

Present readmission risk predictions, LLM-supported discharge guidance, and estimated resource impact through an interactive dashboard, executive memo, and final presentation.

**Execute:**

Support healthcare teams in identifying high-risk patients and targeting post-discharge interventions, while allowing hospitals to adjust the decision threshold based on available resources.

# DELTA Plus Readiness Assessment

**Data:**

UCI Diabetes data + CMS/AHRQ guidance support prediction and discharge planning.

**Enterprise:**

Supports hospital administrators and care coordination teams during discharge planning.

**Leadership:**

Hospital and clinical leadership would oversee real-world implementation.

**Targets:**

Reduce 30-day readmissions while improving care, resource allocation, and costs.

**Analysts:**

Data scientists, healthcare analysts, and clinical experts support development and validation.

**Technology:**

Python, Gradio, Hugging Face Spaces, and an LLM-based RAG system.

**Analytics:**

Predictive modeling, RAG guidance, and cost/value analysis support targeted interventions.

# Data Preparation & Feature Engineering

## Data:

UCI Diabetes 130-US Hospitals

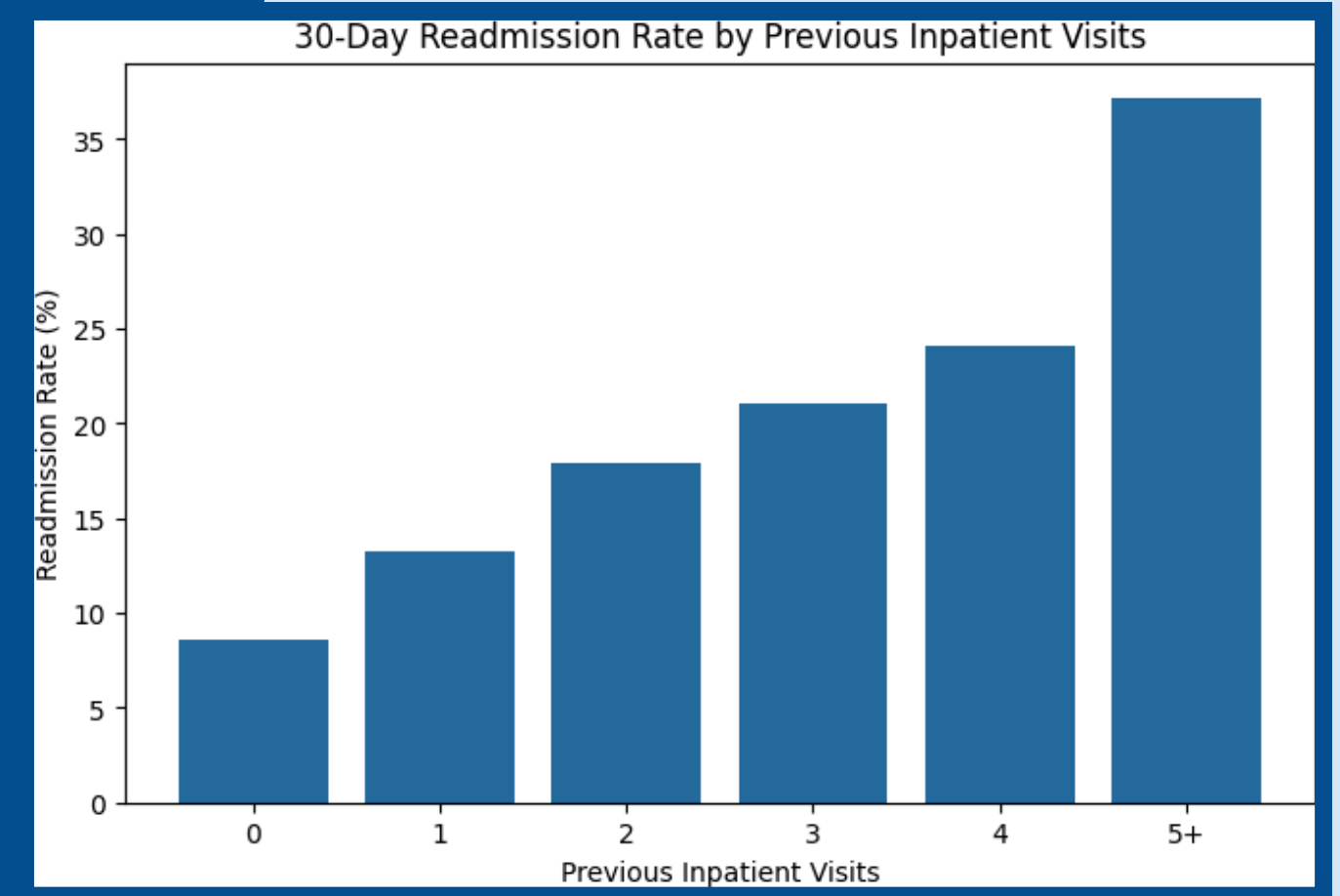
- 101,766 hospital encounters
- 130 U.S. hospitals
- 1999–2008
- Patients with diabetes
- Target: 30-day readmission

## Data Preparation:

- Handled missing and unknown values
- Removed identifiers and non-predictive variables
- Grouped diagnosis codes into broader categories
- Recoded categorical variables for modeling
- Created binary 30-day readmission target
- Reduced to 16 final predictors

## EDA:

- Examined readmission patterns across demographics, hospital utilization, diagnoses, and encounter characteristics
- Prior healthcare utilization showed some of the clearest differences in readmission risk
- Admission/discharge factors and patient complexity also showed meaningful patterns
- EDA helped guide feature selection and model development





# Model Development & Analysis

## Developed Models:

- Logistic Regression
- Random Forest
- XGBoost
- Fine-Tuned LLM

## Analysis Metrics:

- AUROC
- Precision & Recall
- F1 Score
- Confusion Matrix
- Classification threshold analysis
- Cost/resource impact considered when evaluating thresholds

## Model Selection:

- Compared performance across classification approaches
- Tested additional tuning for XGBoost
- Tuning did not provide enough improvement to replace the original model
- Original XGBoost selected as the final predictive model
- Threshold evaluated separately to balance identifying readmissions with the number of patients flagged



# Model Results



| Model               | Threshold | Precision | Recall       | F1-Score | ROC-AUC      |
|---------------------|-----------|-----------|--------------|----------|--------------|
| Logistic Regression | 0.500     | 0.183     | 0.544        | 0.274    | 0.663        |
| Random Forest       | 0.500     | 0.204     | 0.446        | 0.280    | 0.668        |
| <b>XGBoost</b>      | 0.500     | 0.187     | <b>0.557</b> | 0.279    | <b>0.669</b> |
| Fine-Tuned LLM      | 0.422*    | 0.128     | 0.724        | 0.217    | 0.568        |

**ROC-AUC is threshold independent. Precision, recall, and F1 should be interpreted considering the LLM's different selected threshold.**

## Key Findings:

- Traditional ML models performed similarly, with XGBoost achieving the highest ROC-AUC (0.669).
- XGBoost also had the highest recall (0.557) among the traditional models.
- The Fine-Tuned LLM achieved higher recall at its selected threshold, but had lower precision, F1, and ROC-AUC.
- XGBoost was selected as the final model based on its stronger overall performance and practical use for readmission prediction.

# XGBoost Threshold Analysis

*XGBoost was selected as the final model based on its stronger overall performance and highest ROC-AUC.*

| Threshold   | Precision    | Recall       | F1           | Patients Flagged | Readmissions Caught | Missed     |
|-------------|--------------|--------------|--------------|------------------|---------------------|------------|
| 0.30        | 0.122        | 0.953        | 0.216        | 17,677           | 2,154               | 106        |
| <b>0.40</b> | <b>0.148</b> | <b>0.792</b> | <b>0.249</b> | <b>12,103</b>    | <b>1,789</b>        | <b>471</b> |
| 0.50        | 0.187        | 0.557        | 0.279        | 6,749            | 1,259               | 1,001      |
| 0.60        | 0.242        | 0.324        | 0.277        | 3,019            | 732                 | 1,528      |



## Key Takeaways:

- Lower thresholds identify more readmissions but require interventions for more patients.
- 0.40 was selected over 0.50 because recall increased from 55.7% to 79.2%, identifying 530 additional readmissions.
- This prioritizes identifying at-risk patients while avoiding the much larger intervention volume at 0.30.



# RAG-Based Discharge Support



## **Purpose:**

Provide evidence based post-discharge guidance to support care teams after a patient is identified as high risk.

## **Knowledge Base:**

- CMS Hospital Readmissions Reduction Program (HRRP)
- AHRQ Re-Engineered Discharge (RED) Toolkit
- AHRQ RED Toolkit: Tool 5, How to Conduct a Postdischarge Followup Phone Call

## **RAG Process:**

CMS/AHRQ Documents → Document Chunking → Embeddings  
→ Vector Search → Relevant Guidance Retrieved →  
LLM-Generated Response

## Evidence-Based Discharge Support

Use this section to explore ways to support patients after discharge and help reduce their risk of readmission.

Ask a question about **follow-up care, medications, patient education, discharge planning, or care coordination**. The tool searches the available **AHRQ and CMS guidance** and uses the most relevant information to provide practical recommendations.

Recommendations are meant to support discharge planning and should be considered alongside clinical judgment.

Ask a Question

What follow up care is needed for a high risk patient?

Get Discharge Guidance

### Recommendations

#### Recommendation 1

Action: Make a postdischarge follow-up phone call 2 to 3 days after discharge to assess health status, medicines, appointments, and home services.

Why: This helps identify misunderstandings, medication discrepancies, and barriers to follow-up early.

Source: How To Conduct a Postdischarge Follow-up Phone Call, Page: 1

# Cost & Business Impact



| Assumption                 | Value           |
|----------------------------|-----------------|
| XGBoost Threshold          | <b>0.40</b>     |
| Cost per Readmission       | <b>\$16,300</b> |
| Cost per Intervention      | <b>\$200</b>    |
| Intervention Effectiveness | <b>20%</b>      |

*Readmission cost based on AHRQ 2020 data. Intervention cost and effectiveness are adjustable scenario assumptions.*

## Estimated Business Impact:

- 12,103 Patients Flagged
- 1,789 Readmissions Identified
- 358 Potential Readmissions Avoided

**\$3.41M Estimated Net Value**  
**141% Estimated ROI**  
**8.3% Break-Even**

\$5.83M estimated savings  
& \$2.42M intervention cost

## Key Takeaway:

Targeted intervention at the 0.40 threshold could create positive financial value while prioritizing patients at greater risk of readmission.

# Deployed Decision-Support Application

<https://huggingface.co/spaces/chloeprowse/hospital-readmission-decision-support>



# Recommendation & Next Steps

## Recommendation:

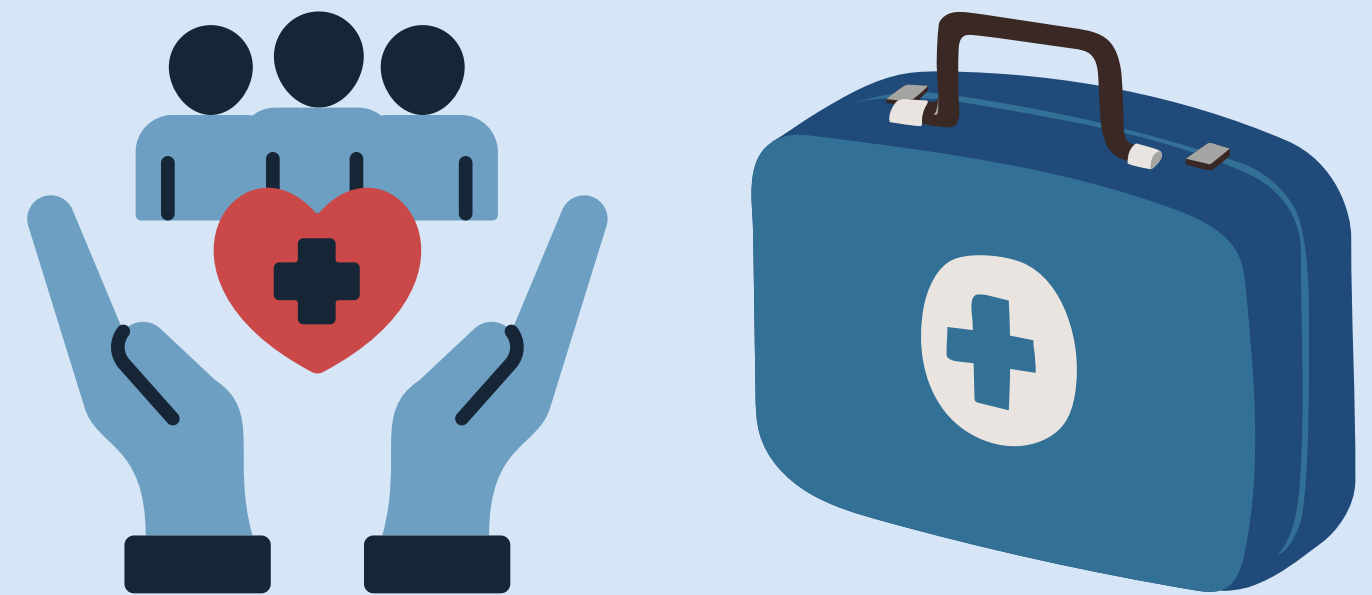
Use XGBoost to identify higher-risk patients for targeted post-discharge support, with the threshold adjusted based on hospital resources and intervention capacity.

## Limitations:

- Historical dataset may not represent current hospital populations
- Model performance is moderate, not sufficient for independent clinical decisions
- Cost results depend on assumed intervention cost and effectiveness
- RAG responses depend on the quality and scope of source documents

## Next Steps:

- Validate the model using current hospital data
- Evaluate performance across patient populations and hospital settings
- Test actual intervention effectiveness and costs
- Expand and regularly update the RAG knowledge base



**This supports clinical decision-making and DOES NOT replace clinical judgment**



# Thank You

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# References

Agency for Healthcare Research and Quality. (2025). Re-Engineered Discharge (RED) Toolkit: Tool 5—How to conduct a postdischarge followup phone call. U.S. Department of Health and Human Services.

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